Benchmarking Enrichment Packages

ORA and MSEA for metabolomics using ad-hoc metabolite sets

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Introduction

Enrichment analysis aims to facilitate the biological interpretation of omics results by relating lists of features (genes, proteins, metabolites) to prior biological knowledge stored in databases (pathways, ontologies, chemical classes, etc.).

In genomics, ORA and GSEA workflows are now fairly standardized. In metabolomics, however, tools are more fragmented and rely on different identifier systems and pathway databases.

The goals of this document are:

1. **Quick-start:** show how to run basic over-representation analysis (ORA) and metabolite set enrichment analysis (MSEA/GSEA-like) with several tools.
2. **Naïve benchmarking:** compare the results of these tools on the same datasets, using MetaboAnalyst Web output as a "practical reference"gold standard".

We will focus on:

- `enrichmet` (pathway enrichment, GSEA, centrality, plots)
- `localEnrichment` (our own infrastructure for metabolite sets and ORA/MSEA)
- `ChemRich` (which does not prohttps://raw.githubusercontent.com/barupal/ChemRICH/refs/heads/master/chemrich_min
- `MetaboAnalystR`, has become very hard to make it work so the examples will be run from its web site and downloaded.

Rather than exhaustively exploring each tool, we restrict ourselves to a reproducible workflow:

1. Use an **ad-hoc collection of metabolite sets**, that is Metabolite sets that have been programatically prepared, as data.frames and `EnrichmentSet` objects, to be used for these analyses..

2. For each dataset, derive:

   - a list of **significant metabolites** (*signature*),
   - a **background** tailored to that signature created either ad-hoc or from the original dataset.

3. Run ORA (and MSEA when possible) with the different tools on the same sets.

Data and metabolite sets

Any pathway analysis requires two inputs:

- A list of features usually associated with a phenotype (e.g. metabolites differentially quantified between two groups).

- A source of annotations for these features, that can be generically described as *Feature sets* or, as in this cas *Metabolite Sets*.

The datasets

We use two datasets that have been preprocessed elsewhere (see file Benchmarking-PrepareData.Rmd).

1. The **Cachexia** dataset. This is a popular example used by MetaboAnalyst, where a comparison between affected and control patients is performed. It has the drawback that only *differential* metabolites are provided, which makes it only partially interesting for some types of analyses. We keep it for its popularity in MetaboAnalyst.

2. The **ST000291** dataset, downloaded from Metabolomics Workbench, and used in a `fobitools` use case. This is obtained by a nutritional intervention study where a comparion among consumption of Apple, Cranberries and Baseline has been performed. In this case we have the full dataset and the results of comparing the groups using limma whih leads to around 40 differential metabolites. We focus only on the *Cranberry vs Baseline* comparison. This study seems to have been made using a cross-over design that the authors have later omitted when providing the samples. Because it is only for instructional purposes, this flaw will be ignored.

Loading datasets

The data for this use case has been prepared elsewhere and is available in the `data`directory.

For each dataset we have the list of metabolites, the analysis results in the form of a topTable and a mapping performed with MetaboAnalyst between the metabolite ids and several databases.

```r
# Cachexia: topTable + ID mapping
cachexia_map_path  <- "data/Cachexia_map.csv"
cachexia_tt_path   <- "data/Cachexia_topTable.Rda"

Cachexia_map <- read.csv(cachexia_map_path, stringsAsFactors = FALSE)
load(cachexia_tt_path)  # loads 'topCachexia'

cat("Cachexia_map:\n")
```

```
## Cachexia_map:
```

```r
print_tbl(head(Cachexia_map))
```

```
## # A tibble: 6 x 9
##   Query                Match     HMDB  PubChem ChEBI KEGG  METLIN SMILES Comment
##   <chr>                <chr>     <chr>   <int> <chr> <chr>  <int> <chr>    <int>
## 1 2-Hydroxyisobutyrate 2-Hydrox~ HMDB~ 4277439 19641 <NA>      NA CC(C)~       1
## 2 Creatinine           Creatini~ HMDB~     588 16737 C007~      8 CN1CC~       1
## 3 Quinolinate          Quinolin~ HMDB~    1066 16675 C037~    330 OC(=O~       1
## 4 Valine               L-Valine  HMDB~    6287 16414 C001~   5842 CC(C)~       1
## 5 Dimethylamine        Dimethyl~ HMDB~     674 17170 C005~   3758 CNC          1
## 6 Tryptophan           L-Trypto~ HMDB~    6305 16828 C000~   5879 N[C@@~       1
```

```r
cat("\ntopCachexia:\n")
```

```
##
## topCachexia:
```

```r
print_tbl(head(topCachexia))
```

```
## # A tibble: 6 x 6
##   groupCachexia groupControl AveExpr     F  P.Value adj.P.Val
##           <dbl>        <dbl>   <dbl> <dbl>    <dbl>     <dbl>
## 1          43.2         27.9    37.3 102.  3.11e-22  1.96e-20
## 2       10722.        5619.   8734.   88.7 1.40e-20  4.41e-19
## 3          83.7         39.3    66.4  85.7 3.48e-20  7.31e-19
## 4          45.6         20.1    35.7  72.8 2.25e-18  3.55e-17
## 5         454.         209.    358.   68.1 1.17e-17  1.48e-16
## 6          81.8         41.8    66.2  65.1 3.56e-17  3.74e-16
```

```r
# ST000291: Cranberry vs Baseline topTable + ID mapping
st291_map_path   <- "data/MW_ST000291__map.csv"
st291_tt_path    <- "data/MW_ST000291-limma_res.Rda"

MS_ST000291_map <- read.csv(st291_map_path, stringsAsFactors = FALSE)
load(st291_tt_path)  # loads 'topCranVSBase'
topCranVSBase<- limma_resCB; rm(limma_resCB)
topCranVSBase_HMDB<- limma_HMDB_CB; rm(limma_HMDB_CB)

cat("\nMS_ST000291_map:\n")
```

```
##
## MS_ST000291_map:
```

```r
print_tbl(head(MS_ST000291_map))
```

```
## # A tibble: 6 x 9
```

```
##      Query Match                  HMDB  PubChem ChEBI KEGG  METLIN SMILES Comment
##      <int> <chr>                  <chr>   <int> <int> <chr>  <int> <chr>    <int>
## 1   192798 Digitalose             <NA>   192798    NA <NA>      NA <NA>         1
## 2    79034 12-Hydroxydodecanoic~ HMDB~    79034 39567 C083~   6465 OCCCC~       1
## 3    10413 4-Hydroxybutyric acid HMDB~    10413 30830 C009~   5678 OCCCC~       1
## 4   439230 Mevalonic acid        HMDB~   439230 17710 C004~    127 C[C@@~       1
## 5 20975673 <NA>                  <NA>        NA    NA <NA>      NA <NA>         0
## 6  5275508 <NA>                  <NA>        NA    NA <NA>      NA <NA>         0
```

```
cat("\ntopCranVSBase:\n")
```

```
##
## topCranVSBase:
```

```
print_tbl(head(topCranVSBase))
```

```
## # A tibble: 6 x 7
##   PubChemCID log2FC   AveExpr     t       pvalue adj_pvalue     B
##   <chr>       <dbl>     <dbl> <dbl>        <dbl>      <dbl> <dbl>
## 1 54678503     2.92  3.57e-16  6.45 0.0000000623  0.0000847  8.11
## 2 71485        3.27 -2.40e-16  5.85 0.000000484   0.000329   6.19
## 3 94214        2.18 -8.93e-17  5.69 0.000000843   0.000382   5.68
## 4 5378303      2.41  1.10e-16  5.54 0.00000140    0.000403   5.20
## 5 3037         3.18  8.39e-17  5.46 0.00000184    0.000403   4.95
## 6 10178        3.33  3.81e-17  5.45 0.00000189    0.000403   4.92
```

```
cat("\ntopCranVSBase (HMDB):\n")
```

```
##
## topCranVSBase (HMDB):
```

```
print_tbl(head(topCranVSBase_HMDB))
```

```
## # A tibble: 6 x 7
##   feature      log2FC   AveExpr     t     pvalue adj_pvalue     B
##   <chr>         <dbl>     <dbl> <dbl>      <dbl>      <dbl> <dbl>
## 1 HMDB0251200   3.18   8.39e-17  5.48 0.00000172   0.000876  5.01
## 2 HMDB0248806   2.26   3.85e-17  5.34 0.00000281   0.000876  4.54
## 3 HMDB0258615   2.05   2.68e-16  5.00 0.00000881   0.00154   3.47
## 4 HMDB0006115   1.56  -4.28e-16  4.96 0.00000986   0.00154   3.36
## 5 HMDB0000738   1.46   1.01e-15  4.72 0.0000221    0.00233   2.60
## 6 HMDB0000714   1.51  -8.74e-16  4.72 0.0000225    0.00233   2.58
```

Metabolite Sets

We also use a **unified collection of metabolite sets**, previously constructed and saved as:

- `KEGGset_HMDB`, `KEGGset_HMDB_df`
- `KEGGset_PubChem`, `KEGGset_PubChem_df`
- `ChemicalClassSet`, `ChemicalClassSet_df`
- `SMPDBset`, `SMPDBset_df`

All sets collections, but `KEGGset_PubChem` use **HMDB IDs** as feature identifiers. `KEGGset_PubChem` uses **PubChemIDs**.

Load Metabolite Sets

Metabolite sets have been prepared elsewhere and saved as `EnrichmentSet` objects of the class defined in the `localEnrichment` package.

```r
# Metabolite sets (EnrichmentSet collection)
# mets_collection_path <- "databases/MetaboliteSets_Collection.Rda"
# load(mets_collection_path)

load("C:/Users/Usuario/OneDrive - Universitat de Barcelona/EstBioinfo/Databases4Metabolomics/results/Me

### Parche. Pendent de netejar-ho al script  que ho crea
KEGGset_HMDB_df <- df_HMDB
KEGGset_PubChem_df <- df_PubChem
rm(df_HMDB, df_PubChem, KEGGset_global, KEGGset_global_df, KEGGset_study, KEGGset_study_df)
###

cat("\nEnrichmentSet objects loaded:\n")
```

```
##
## EnrichmentSet objects loaded:
```

```r
summary(KEGGset_HMDB);      cat("\n")
```

```
## EnrichmentSet summary:
##   Mapping name: KEGG_pathway_HMDB
##   Source: KEGG
##   Feature IDs: HMDB
##   Number of sets: 282
##   Mean set size: 22.60993 features
##   Median set size: 9.5 features
```

```r
summary(KEGGset_PubChem);      cat("\n")
```

```
## EnrichmentSet summary:
##   Mapping name: KEGG_pathway_PubChem
##   Source: KEGG
##   Feature IDs: PubChem
##   Number of sets: 266
##   Mean set size: 18.18045 features
##   Median set size: 8 features
```

```r
summary(ChemicalClassSet);    cat("\n")
```

```
## EnrichmentSet summary:
##   Mapping name: ChemicalClasses
##   Source: HMDB
##   Feature IDs: HMDB
##   Number of sets: 30
##   Mean set size: 9.366667 features
##   Median set size: 3.5 features
```

```r
summary(SMPDBset);            cat("\n")
```

```
## EnrichmentSet summary:
##   Mapping name: SMPDB_pathways
##   Source: SMPDB
##   Feature IDs: HMDB
```

```
##   Number of sets: 48654
##   Mean set size: 17.30571 features
##   Median set size: 18 features
```

```
cat("\nKEGGset_HMDB_df:\n")
```

```
##
## KEGGset_HMDB_df:
```

```
print_tbl(head(KEGGset_HMDB_df))
```

```
## # A tibble: 6 x 3
##   set_name                                                    set_id Metabolites
##   <chr>                                                       <chr>  <chr>
## 1 Glycolysis / Gluconeogenesis - Homo sapiens (human)         hsa00~ HMDB000024~
## 2 Citrate cycle (TCA cycle) - Homo sapiens (human)            hsa00~ HMDB000024~
## 3 Pentose phosphate pathway - Homo sapiens (human)            hsa00~ HMDB000024~
## 4 Pentose and glucuronate interconversions - Homo sapiens (h~ hsa00~ HMDB000024~
## 5 Fructose and mannose metabolism - Homo sapiens (human)      hsa00~ HMDB000012~
## 6 Galactose metabolism - Homo sapiens (human)                 hsa00~ HMDB000028~
```

```
cat("\nKEGGset_PubChem_df:\n")
```

```
##
## KEGGset_PubChem_df:
```

```
print_tbl(head(KEGGset_PubChem_df))
```

```
## # A tibble: 6 x 3
##   set_name                                                    set_id Metabolites
##   <chr>                                                       <chr>  <chr>
## 1 Glycolysis / Gluconeogenesis - Homo sapiens (human)         hsa00~ 1060;10773~
## 2 Citrate cycle (TCA cycle) - Homo sapiens (human)            hsa00~ 1060;10773~
## 3 Pentose phosphate pathway - Homo sapiens (human)            hsa00~ 1060;10773~
## 4 Pentose and glucuronate interconversions - Homo sapiens (h~ hsa00~ 1060;10773~
## 5 Fructose and mannose metabolism - Homo sapiens (human)      hsa00~ 439709;668~
## 6 Galactose metabolism - Homo sapiens (human)                 hsa00~ 8629;5988;~
```

```
cat("\nChemicalClassSet_df:\n")
```

```
##
## ChemicalClassSet_df:
```

```
print_tbl(head(ChemicalClassSet_df))
```

```
## # A tibble: 6 x 3
##   set_name                 set_id                   Metabolites
##   <chr>                    <chr>                    <chr>
## 1 Acylcarnitines           Acylcarnitines           HMDB0000201;HMDB0000824;HMDB0~
## 2 Amine oxide              Amine oxide              HMDB0000925
## 3 Amino Acids              Amino Acids              HMDB0000187;HMDB0000687;HMDB0~
## 4 Amino Acids Derivatives Amino Acids Derivatives HMDB0000092;HMDB0000812;HMDB0~
## 5 Amino acid - related     Amino acid - related     HMDB0000452;HMDB0000510;HMDB0~
## 6 Azoles                   Azoles                   HMDB0000462
```

```
cat("\nSMPDBset_df:\n")
```

```
##
## SMPDBset_df:
```

```r
print_tbl(head(SMPDBset_df))
```

```
## # A tibble: 6 x 3
##   set_name                                 set_id     Metabolites
##   <chr>                                    <chr>      <chr>
## 1 Citrullinemia Type I                     SMP0000001 HMDB0000641;HMDB0000161;~
## 2 Carbamoyl Phosphate Synthetase Deficiency SMP0000002 HMDB0000641;HMDB0000161;~
## 3 Argininosuccinic Aciduria                SMP0000003 HMDB0000641;HMDB0000161;~
## 4 Glycine and Serine Metabolism            SMP0000004 HMDB0002134;HMDB0001377;~
## 5 Pterine Biosynthesis                     SMP0000005 HMDB0006822;HMDB0002111;~
## 6 Tyrosine Metabolism                      SMP0000006 HMDB0000158;HMDB0000208;~
```

Building signatures and backgrounds

Our aim is to derive **signatures** and **backgrounds** that are compatible with the `EnrichmentSet` objects, which use **HMDB IDs**.

We therefore:

1. Map each topTable to HMDB IDs.
2. Construct a vector of significant HMDB IDs (signature).
3. Use `filterEnrichmentSet(Eset, ids)` to restrict the metabolite sets to those that contain at least one of the signature metabolites.
4. Build a **background** as the union of all metabolites in the filtered sets.

Cachexia: HMDB mapping and signature

```r
library(tibble)
```

```
## Warning: package 'tibble' was built under R version 4.4.3
```

```r
# Convert topCachexia to tibble with metabolite name
cachexia_tt <- topCachexia %>%
  rownames_to_column(var = "Metabolite") %>%
  left_join(Cachexia_map, by = c("Metabolite" = "Query"))

cat("Cachexia limma table with mapping:\n")
```

```
## Cachexia limma table with mapping:
```

```r
print_tbl(cachexia_tt)
```

```
## # A tibble: 63 x 15
##   Metabolite   groupCachexia groupControl AveExpr     F  P.Value adj.P.Val Match
##   <chr>                <dbl>        <dbl>   <dbl> <dbl>    <dbl>     <dbl> <chr>
## 1 2-Hydroxyis~          43.2         27.9    37.3 102.  3.11e-22  1.96e-20 2-Hy~
## 2 Creatinine         10722.        5619.   8734.   88.7 1.40e-20  4.41e-19 Crea~
## 3 Quinolinate           83.7         39.3    66.4  85.7 3.48e-20  7.31e-19 Quin~
## 4 Valine                45.6         20.1    35.7  72.8 2.25e-18  3.55e-17 L-Va~
## 5 Dimethylami~         454.         209.    358.   68.1 1.17e-17  1.48e-16 Dime~
## 6 Tryptophan            81.8         41.8    66.2  65.1 3.56e-17  3.74e-16 L-Tr~
## # i 57 more rows
## # i 7 more variables: HMDB <chr>, PubChem <int>, ChEBI <chr>, KEGG <chr>,
## #   METLIN <int>, SMILES <chr>, Comment <int>
```

```r
# p-value column and threshold
p_col_cachexia <- "adj.P.Val"
alpha_cachexia <- 0.05

# Signature: HMDB IDs of significant metabolites
cachexia_sig <- cachexia_tt %>%
  filter(!is.na(HMDB),
         !!rlang::sym(p_col_cachexia) < alpha_cachexia) %>%
  pull(HMDB) %>%
  unique()

length(cachexia_sig)
```

```
## [1] 61
```

```r
cachexia_sig
```

```
##  [1] "HMDB0242161" "HMDB0000562" "HMDB0000232" "HMDB0000883" "HMDB0000087"
##  [6] "HMDB0000929" "HMDB0000164" "HMDB0000267" "HMDB0000687" "HMDB0000168"
## [11] "HMDB0000092" "HMDB0000479" "HMDB0000167" "HMDB0000043" "HMDB0000161"
## [16] "HMDB0000682" "HMDB0000174" "HMDB0000641" "HMDB0000187" "HMDB0000149"
## [21] "HMDB0000094" "HMDB0000958" "HMDB0000157" "HMDB0000128" "HMDB0000158"
## [26] "HMDB0000172" "HMDB0000177" "HMDB0000300" "HMDB0000123" "HMDB0000754"
## [31] "HMDB0000042" "HMDB0000020" "HMDB0000211" "HMDB0000243" "HMDB0000182"
## [36] "HMDB0000714" "HMDB0304356" "HMDB0000640" "HMDB0000072" "HMDB0000251"
## [41] "HMDB0000254" "HMDB0000925" "HMDB0000875" "HMDB0001522" "HMDB0000062"
## [46] "HMDB0000001" "HMDB0000452" "HMDB0000201" "HMDB0000134" "HMDB0000448"
## [51] "HMDB0000699" "HMDB0000064" "HMDB0000210" "HMDB0001659" "HMDB0000122"
## [56] "HMDB0000258" "HMDB0000208" "HMDB0003911" "HMDB0000098" "HMDB0000190"
## [61] "HMDB0000956"
```

Note that in this Cachexia dataset we only have *differential* metabolites, not the full list of measured features. A realistic background based purely on the data is therefore impossible. Instead, we derive a *functional background* from the metabolite-set collection.

The background will be constructed artifically by taking:

- All sets in the `KEGGset_HMDB` collection.
- The HMDB ids of each set.

```r
# Extract background as the union of all HMDB IDs in the filtered sets

cachexia_background <- unique(unlist(KEGGset_HMDB@data$feature_list))
length(cachexia_background)
```

```
## [1] 2106
```

```r
intersect(cachexia_sig, cachexia_background) |> length()
```

```
## [1] 45
```

We now have all we need to do the Enrichment Analysis on the Cachexia dataset.

- The list of *significant* metabolites: `cachexia_sig`

```r
str(cachexia_sig)
```

```
##  chr [1:61] "HMDB0242161" "HMDB0000562" "HMDB0000232" "HMDB0000883" ...
```

- The *background* associated with this list: `cachexia_background`

```r
str(cachexia_background)
```

```
##  chr [1:2106] "HMDB0000243" "HMDB0001206" "HMDB0000042" "HMDB0000223" ...
```

- The collections of metabolite sets to do the enrichment Analysis: `KEGGset_HMDB`.

```r
summary(KEGGset_HMDB)
```

```
## EnrichmentSet summary:
##   Mapping name: KEGG_pathway_HMDB
##   Source: KEGG
##   Feature IDs: HMDB
##   Number of sets: 282
##   Mean set size: 22.60993 features
##   Median set size: 9.5 features
```

- We can also use `SMPDBset` and `ChemicalClassSet` because these are sets made of HMDB ids.

```r
summary(SMPDBset)
```

```
## EnrichmentSet summary:
##   Mapping name: SMPDB_pathways
##   Source: SMPDB
##   Feature IDs: HMDB
##   Number of sets: 48654
##   Mean set size: 17.30571 features
##   Median set size: 18 features
```

```r
summary(ChemicalClassSet)
```

```
## EnrichmentSet summary:
##   Mapping name: ChemicalClasses
##   Source: HMDB
##   Feature IDs: HMDB
##   Number of sets: 30
##   Mean set size: 9.366667 features
##   Median set size: 3.5 features
```

ST000291: PubChem and HMDB mappings and signature

ST000291 with PubChem identifiers

The original ST000291 dataset is annotated with `PubChem` identifiers, so we are restricted to work Metabolite Sets created with PubChem identifiers.

The code below shows how to prepare a list of selected metabolites and background straight from the available information.

To define the signature we select metabolites with an adjusted p-value of at most 0.33

```r
## ST000291: signature (PubChem) and background

# top table from limma
st291_tt <- topCranVSBase   # ja carregat anteriorment

# signature (adj.P.Val < 0.05)
ST291_sig <- st291_tt %>%
  filter(pvalue  < 0.05) %>%
  pull(PubChemCID) %>%
```

```r
  unique()

# background = ALL tested metabolites
ST291_background <- st291_tt %>%
  pull(PubChemCID) %>%
  unique()

cat("Signature size:", length(ST291_sig), "\n")
```

```
## Signature size: 76
```

```r
cat("Background size:", length(ST291_background), "\n")
```

```
## Background size: 1358
```

ST000291 with HMDB identifiers

Given that many programs can only work with HMDB or KEGG identifiers we have prepared and analyzed a subset of metabolites made only by those that have such ids.

```r
## ST000291: signature (HMDB) and background

# top table from limma
st291_tt_HMDB <- topCranVSBase_HMDB   # ja carregat anteriorment

# signature (adj.P.Val < 0.05)
ST291_HMDB_sig <- st291_tt_HMDB %>%
  filter(pvalue  < 0.05) %>%
  pull(feature) %>%
  unique()

# background = ALL tested metabolites
ST291_HMDB_background <- st291_tt_HMDB %>%
  pull(feature) %>%
  unique()

cat("Signature size:", length(ST291_HMDB_sig), "\n")
```

```
## Signature size: 29
```

```r
cat("Background size:", length(ST291_HMDB_background), "\n")
```

```
## Background size: 623
```

We now have all we need to do the Enrichment Analysis on the Cachexia dataset.

Enrichment with `localEnrichment`

Overrepresentation Analysis (ORA) for the Cachexia dataset

ORA based on KEGGsets_HMDB

We start doing ORA using KEGG metabolite sets with HMDB identifiers

First perform Fisher test for all sets and select the enriched ones.

```r
library(localEnrichment)
```

```r
# ORA per Cachexia vs KEGG (HMDB)

ORA_cachexia_KEGG <- ora_test(
  eset       = KEGGset_HMDB,
  selected   = cachexia_sig,
  background = cachexia_background,
  test       = "fisher",
  p_adjust   = "BH",
  min_set_size = 3,
  max_set_size = 500
)

# veure com és el resultat
dplyr::glimpse(ORA_cachexia_KEGG)
```

```
## Rows: 99
## Columns: 8
## $ set_id            <chr> "hsa05230", "hsa00970", "hsa04974", "hsa04978", "hsa~
## $ set_name          <chr> "Central carbon metabolism in cancer - Homo sapiens ~
## $ n_selected_in_set <int> 17, 13, 14, 10, 14, 8, 10, 6, 9, 8, 6, 5, 5, 5, 5, 3~
## $ n_in_set          <int> 35, 23, 47, 27, 82, 26, 47, 15, 45, 40, 22, 17, 18, ~
## $ p_value           <dbl> 2.039174e-18, 2.441065e-15, 1.026504e-11, 1.122660e-~
## $ overlap_features  <chr> "HMDB0000883;HMDB0000929;HMDB0000687;HMDB0000168;HMD~
## $ p_adj             <dbl> 2.018783e-16, 1.208327e-13, 3.387462e-10, 2.778584e-~
## $ fold_enrichment   <dbl> 16.769087, 19.513899, 10.283920, 12.786885, 5.894442~
```

```r
head(ORA_cachexia_KEGG)
```

```
## # A tibble: 6 x 8
##   set_id  set_name n_selected_in_set n_in_set  p_value overlap_features    p_adj
##   <chr>   <chr>                <int>    <int>    <dbl> <chr>               <dbl>
## 1 hsa052~ Central~                17       35 2.04e-18 HMDB0000883;HMD~ 2.02e-16
## 2 hsa009~ Aminoac~                13       23 2.44e-15 HMDB0000883;HMD~ 1.21e-13
## 3 hsa049~ Protein~                14       47 1.03e-11 HMDB0000883;HMD~ 3.39e-10
## 4 hsa049~ Mineral~                10       27 1.12e- 9 HMDB0000883;HMD~ 2.78e- 8
## 5 hsa020~ ABC tra~                14       82 3.11e- 8 HMDB0000883;HMD~ 6.15e- 7
## 6 hsa002~ Alanine~                 8       26 3.23e- 7 HMDB0000168;HMD~ 5.32e- 6
## # i 1 more variable: fold_enrichment <dbl>
```

```r
ORA_cachexia_KEGG_sig <- ORA_cachexia_KEGG %>%
  dplyr::filter(p_adj < 0.05) %>%
  dplyr::arrange(p_adj)

nrow(ORA_cachexia_KEGG_sig)
```

```
## [1] 26
```

```r
head(ORA_cachexia_KEGG_sig, 10)
```

```
## # A tibble: 10 x 8
##    set_id set_name n_selected_in_set n_in_set  p_value overlap_features    p_adj
##    <chr>  <chr>                <int>    <int>    <dbl> <chr>               <dbl>
##  1 hsa05~ Central~                17       35 2.04e-18 HMDB0000883;HMD~ 2.02e-16
##  2 hsa00~ Aminoac~                13       23 2.44e-15 HMDB0000883;HMD~ 1.21e-13
##  3 hsa04~ Protein~                14       47 1.03e-11 HMDB0000883;HMD~ 3.39e-10
##  4 hsa04~ Mineral~                10       27 1.12e- 9 HMDB0000883;HMD~ 2.78e- 8
```

```
##  5 hsa02~ ABC tra~                 14        82 3.11e- 8 HMDB0000883;HMD~ 6.15e- 7
##  6 hsa00~ Alanine~                  8        26 3.23e- 7 HMDB0000168;HMD~ 5.32e- 6
##  7 hsa00~ Glyoxyl~                 10        47 4.39e- 7 HMDB0000641;HMD~ 6.21e- 6
##  8 hsa00~ Citrate~                  6        15 1.88e- 6 HMDB0000094;HMD~ 2.33e- 5
##  9 hsa00~ D-Amino~                  9        45 3.08e- 6 HMDB0000167;HMD~ 3.39e- 5
## 10 hsa00~ Glycine~                  8        40 1.15e- 5 HMDB0000929;HMD~ 1.14e- 4
## # i 1 more variable: fold_enrichment <dbl>
```

```r
str(ORA_cachexia_KEGG_sig)
```

```
## EnrchmnR [26 x 8] (S3: EnrichmentResult/tbl_df/tbl/data.frame)
##  $ set_id           : chr [1:26] "hsa05230" "hsa00970" "hsa04974" "hsa04978" ...
##  $ set_name         : chr [1:26] "Central carbon metabolism in cancer - Homo sapiens (human)" "Aminoa
##  $ n_selected_in_set: int [1:26] 17 13 14 10 14 8 10 6 9 8 ...
##  $ n_in_set         : int [1:26] 35 23 47 27 82 26 47 15 45 40 ...
##  $ p_value          : num [1:26] 2.04e-18 2.44e-15 1.03e-11 1.12e-09 3.11e-08 ...
##  $ overlap_features : chr [1:26] "HMDB0000883;HMDB0000929;HMDB0000687;HMDB0000168;HMDB0000161;HMDB000
##  $ p_adj            : num [1:26] 2.02e-16 1.21e-13 3.39e-10 2.78e-08 6.15e-07 ...
##  $ fold_enrichment  : num [1:26] 16.77 19.51 10.28 12.79 5.89 ...
##  - attr(*, "metadata")=List of 6
##   ..$ mapping_name   : chr "KEGG_pathway_HMDB"
##   ..$ feature_id_type: chr "HMDB"
##   ..$ feature_species: chr "Homo sapiens"
##   ..$ set_source     : chr "KEGG"
##   ..$ version        : chr "2025-11-29"
##   ..$ description    : chr "KEGG pathway to HMDB metabolites via metaboliteIDmapping"
```

This can be visualized by different approaches.

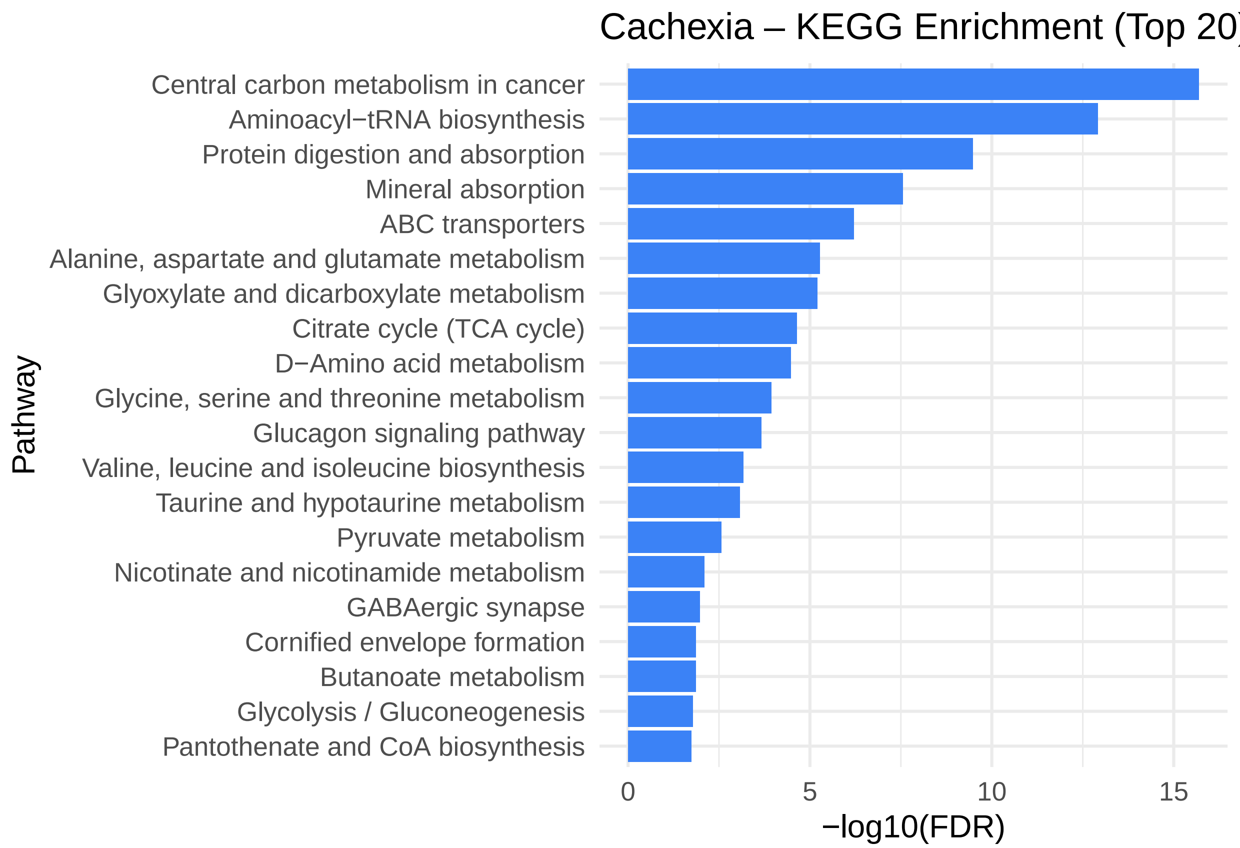
First clean the Pathway names

```r
library(ggplot2)
```

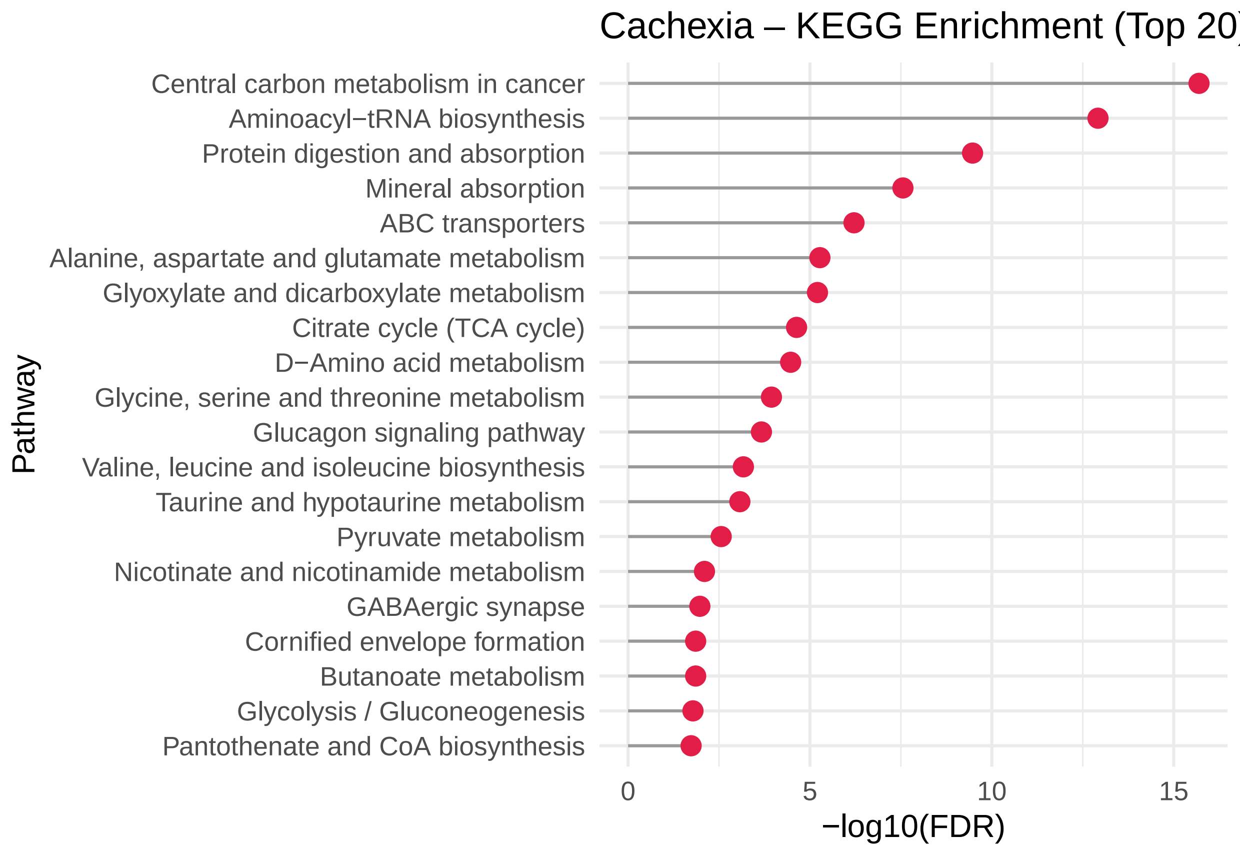
```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```r
clean_pathway_name <- function(x) {
  gsub(" - Homo sapiens \\(human\\)$", "", x, perl = TRUE) |>
    trimws()
}

ORA_cachexia_KEGG_sig <- ORA_cachexia_KEGG_sig %>%
  dplyr::mutate(set_name_clean = clean_pathway_name(set_name))

top20 <- ORA_cachexia_KEGG_sig %>%
  arrange(p_adj) %>%
  slice(1:20)


# IMPORTANT: nivells únics per ordenar el gràfic
top20$set_name_clean <- factor(
  top20$set_name_clean,
  levels = rev(unique(top20$set_name_clean))
)


ggplot(top20, aes(x = set_name_clean, y = -log10(p_adj))) +
  geom_col(fill = "#3B82F6") +
```

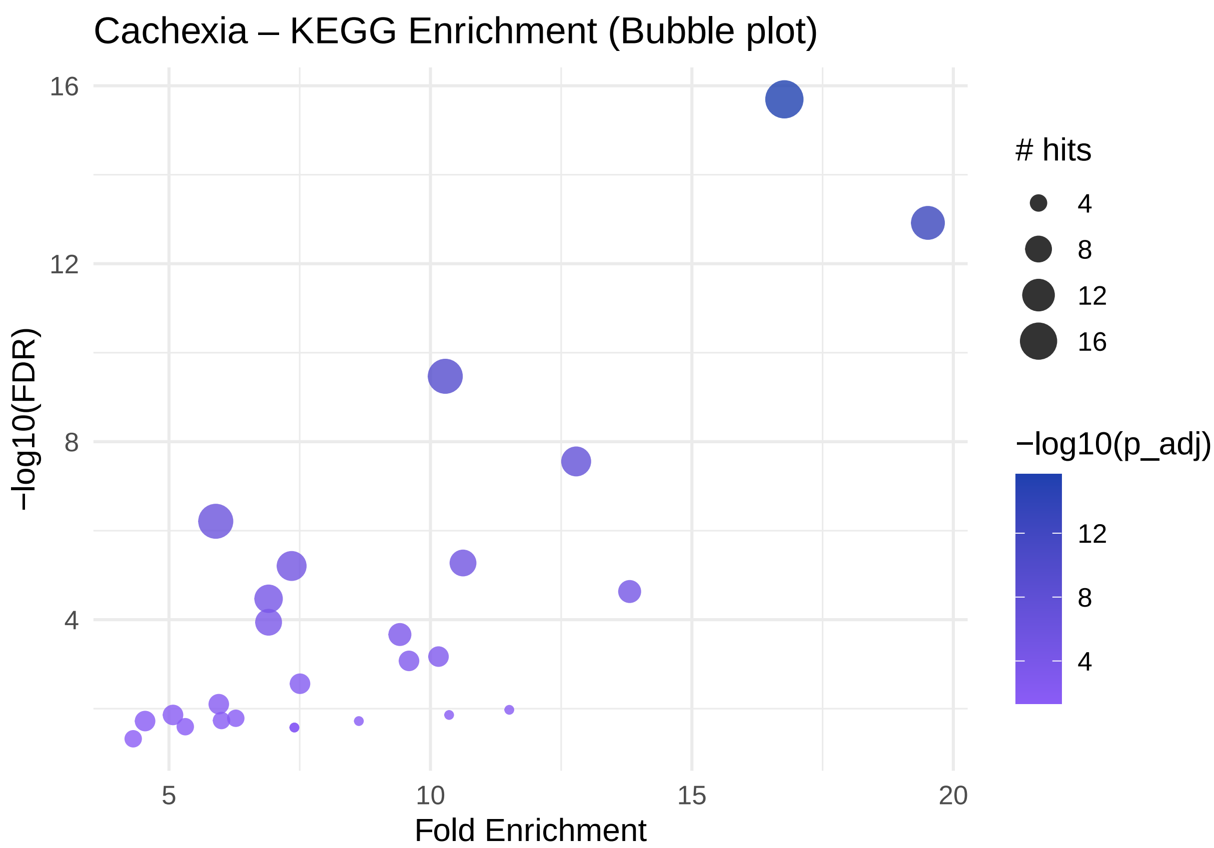
```r
  coord_flip() +
  labs(
    title = "Cachexia - KEGG Enrichment (Top 20)",
    x = "Pathway",
    y = "-log10(FDR)"
  ) +
  theme_minimal(base_size = 12)
```




```r
ggplot(top20, aes(x = -log10(p_adj), y = set_name_clean)) +
  geom_segment(aes(x = 0, xend = -log10(p_adj),
                   y = set_name_clean, yend = set_name_clean),
               color = "#999999") +
  geom_point(size = 3, color = "#E11D48") +
  labs(
    title = "Cachexia - KEGG Enrichment (Top 20)",
    x = "-log10(FDR)",
    y = "Pathway"
  ) +
  theme_minimal(base_size = 12)
```




```r
top30 <- ORA_cachexia_KEGG_sig %>%
  arrange(p_adj) %>%
  slice(1:30)

ggplot(top30, aes(
  x = fold_enrichment,
  y = -log10(p_adj),
  size = n_selected_in_set,
  color = -log10(p_adj)
)) +
  geom_point(alpha = 0.8) +
  scale_color_gradient(low = "#8B5CF6", high = "#1E40AF") +
  labs(
    title = "Cachexia - KEGG Enrichment (Bubble plot)",
    x = "Fold Enrichment",
    y = "-log10(FDR)",
    size = "# hits"
  ) +
  theme_minimal(base_size = 12)
```



ORA based on SMPDBsets

We can proceed similarly with SMPDB

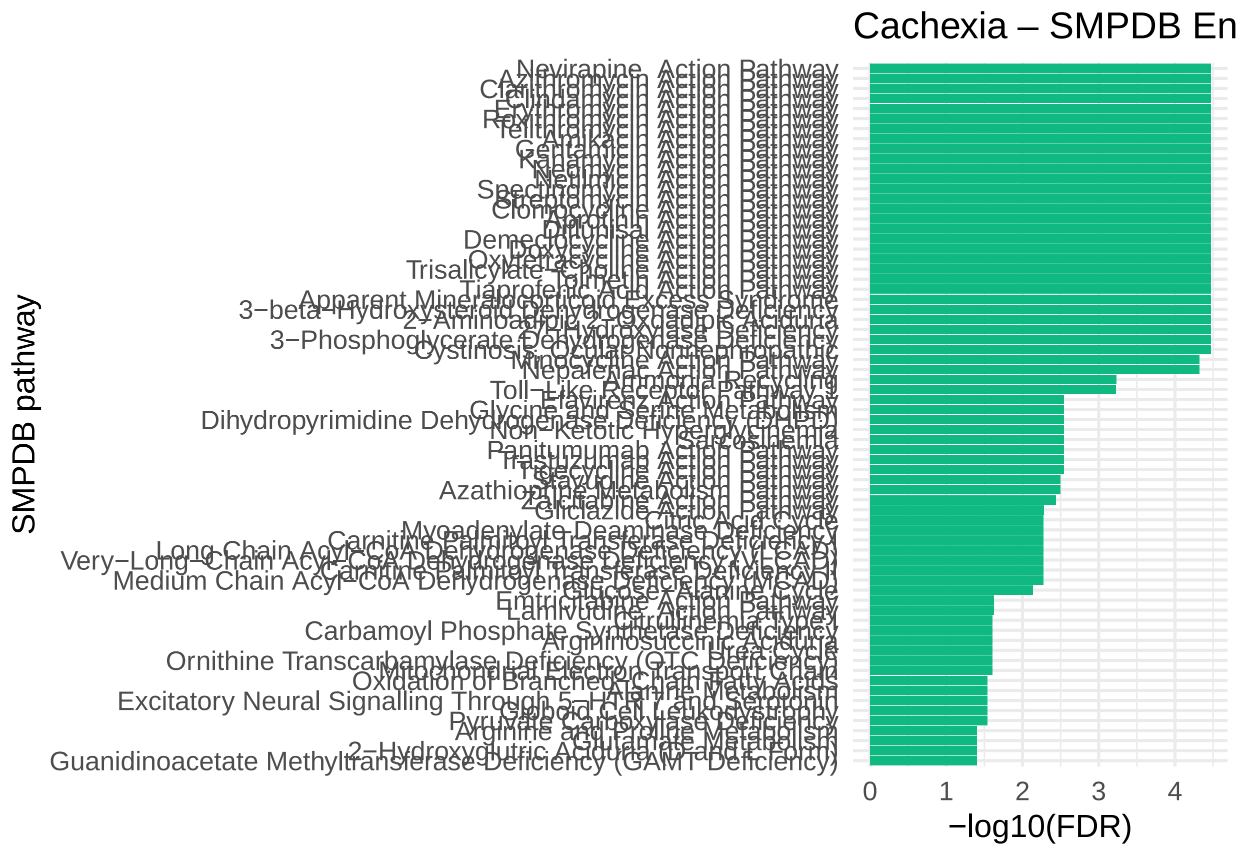
```
# ORA Cachexia vs SMPDB

ORA_cachexia_SMPDB <- ora_test(
  eset          = SMPDBset,
  selected      = cachexia_sig,
  background    = cachexia_background,
  test          = "fisher",
  p_adjust      = "BH",
  min_set_size = 3,
  max_set_size = 500
)


# Resultats significatius
ORA_cachexia_SMPDB_sig <- ORA_cachexia_SMPDB %>%
  filter(p_adj < 0.05) %>%
  arrange(p_adj)

nrow(ORA_cachexia_SMPDB_sig)
```

```
## [1] 87
```

```r
head(ORA_cachexia_SMPDB_sig, 10)
```

```
## # A tibble: 10 x 8
##    set_id   set_name n_selected_in_set n_in_set p_value overlap_features   p_adj
##    <chr>    <chr>                <int>    <int>   <dbl> <chr>              <dbl>
##  1 SMP0000~ Nevirap~                10       41 1.09e-7 HMDB0000687;HMD~ 3.42e-5
##  2 SMP0000~ Azithro~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
##  3 SMP0000~ Clarith~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
##  4 SMP0000~ Clindam~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
##  5 SMP0000~ Erythro~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
##  6 SMP0000~ Roxithr~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
##  7 SMP0000~ Telithr~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
##  8 SMP0000~ Amikaci~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
##  9 SMP0000~ Gentami~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
## 10 SMP0000~ Kanamyc~                 7       19 4.63e-7 HMDB0000883;HMD~ 3.42e-5
## # i 1 more variable: fold_enrichment <dbl>
```

```r
top_smpdb <- ORA_cachexia_SMPDB_sig %>%
  slice(1:70) %>%
  mutate(set_name = factor(set_name, levels = rev(set_name)))


ggplot(top_smpdb, aes(x = set_name, y = -log10(p_adj))) +
  geom_col(fill = "#10B981") +   # color verd (diferent del KEGG)
  coord_flip() +
  labs(
    title = "Cachexia - SMPDB Enrichment (Top 20)",
    x = "SMPDB pathway",
    y = "-log10(FDR)"
  ) +
  theme_minimal(base_size = 12)
```



ORA for chemical classes

```r
ORA_cachexia_Chem <- ora_test(
  eset          = ChemicalClassSet,
  selected      = cachexia_sig,
  background    = cachexia_background,
  test          = "fisher",
  p_adjust      = "BH",
  min_set_size = 1,
  max_set_size = 500
)

ORA_cachexia_Chem_sig <- ORA_cachexia_Chem %>%
  filter(p_adj < 0.25) %>%
  arrange(p_adj)

nrow(ORA_cachexia_Chem_sig)
```

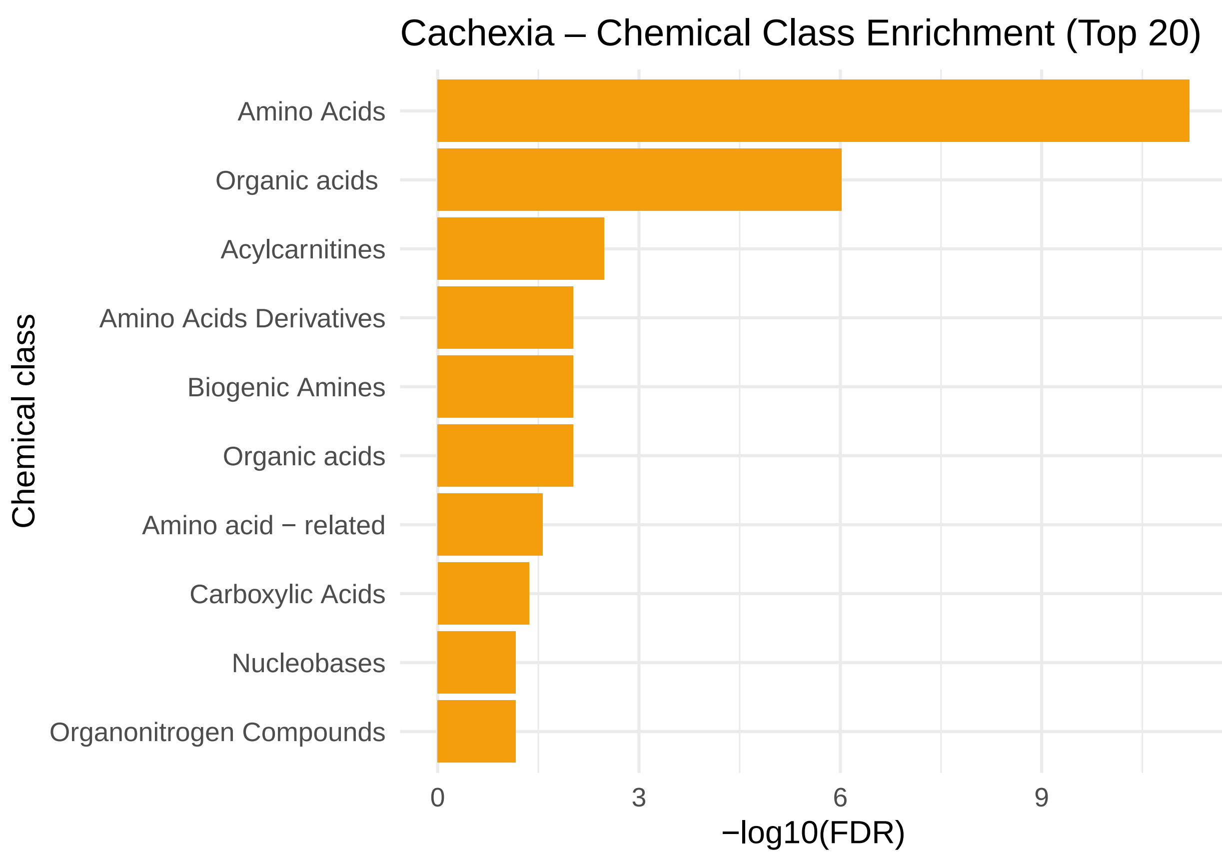
```
## [1] 12
```

```r
head(ORA_cachexia_Chem_sig, 10)
```

```
## # A tibble: 10 x 8
##    set_id set_name n_selected_in_set n_in_set  p_value overlap_features    p_adj
##    <chr>  <chr>                <int>    <int>    <dbl> <chr>               <dbl>
##  1 "Amin~ "Amino ~                10       15 5.23e-13 HMDB0000687;HMD~ 6.28e-12
```

```
##  2 "Orga~ "Organi~                  8       24 1.59e- 7 HMDB0000232;HMD~ 9.54e- 7
##  3 "Acyl~ "Acylca~                  2        2 8.26e- 4 HMDB0000062;HMD~ 3.30e- 3
##  4 "Amin~ "Amino ~                  2        4 4.77e- 3 HMDB0000267;HMD~ 9.54e- 3
##  5 "Biog~ "Biogen~                  2        4 4.77e- 3 HMDB0000562;HMD~ 9.54e- 3
##  6 "Orga~ "Organi~                  2        4 4.77e- 3 HMDB0000128;HMD~ 9.54e- 3
##  7 "Amin~ "Amino ~                  2        7 1.58e- 2 HMDB0000479;HMD~ 2.71e- 2
##  8 "Carb~ "Carbox~                  1        1 2.90e- 2 HMDB0000072      4.34e- 2
##  9 "Nucl~ "Nucleo~                  1        2 5.71e- 2 HMDB0000157      6.85e- 2
## 10 "Orga~ "Organo~                  1        2 5.71e- 2 HMDB0000149      6.85e- 2
## # i 1 more variable: fold_enrichment <dbl>
```

```
top_chem <- ORA_cachexia_Chem_sig %>%
  slice(1:10) %>%
  mutate(set_name = factor(set_name, levels = rev(set_name)))

ggplot(top_chem, aes(x = set_name, y = -log10(p_adj))) +
  geom_col(fill = "#F59E0B") +    # taronja per diferenciar
  coord_flip() +
  labs(
    title = "Cachexia - Chemical Class Enrichment (Top 20)",
    x = "Chemical class",
    y = "-log10(FDR)"
  ) +
  theme_minimal(base_size = 12)
```



ORA and MSEA for the ST000291 dataset

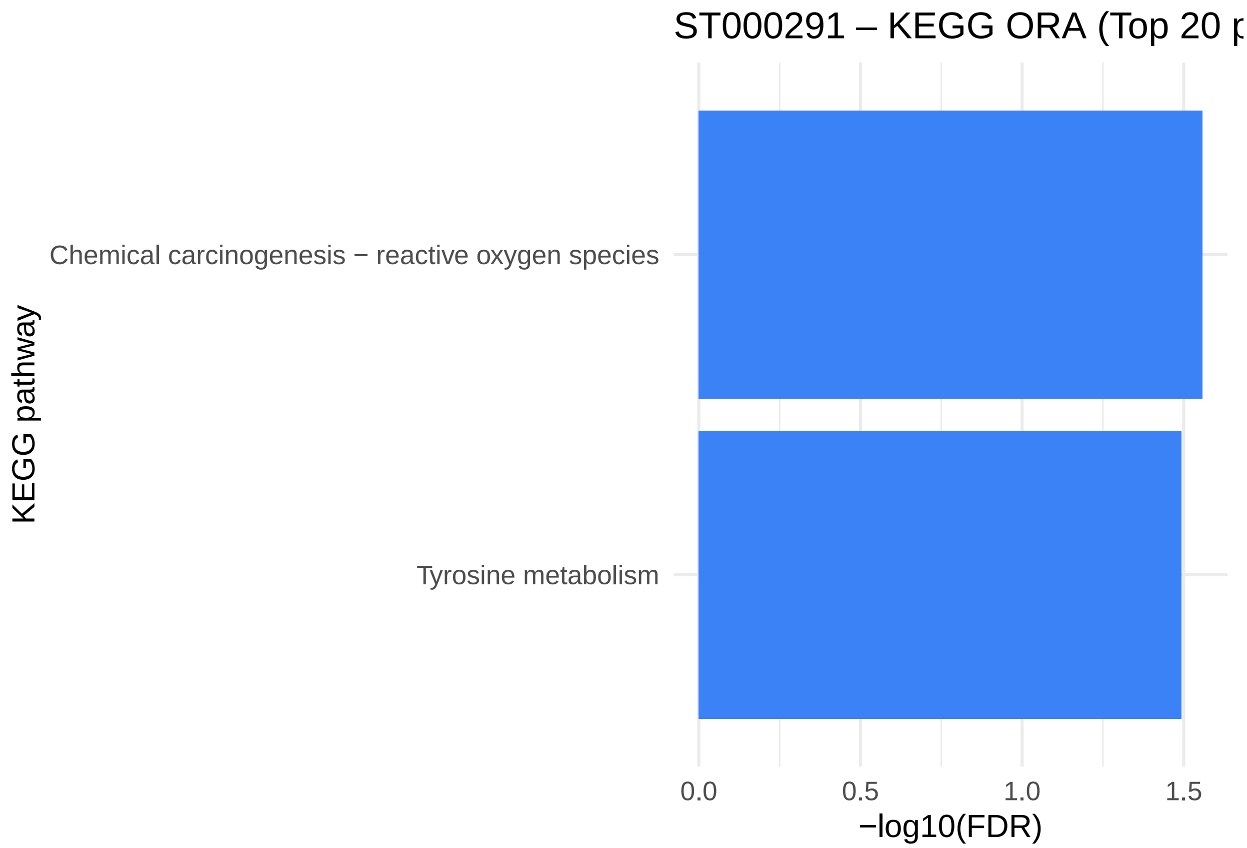
ORA for the ST000291 dataset

```r
## --------------------------------
## 1) ORA (Over-Representation Analysis)
## --------------------------------

ORA_ST291_KEGG <- ora_test(
  eset         = KEGGset_PubChem,
  selected     = ST291_sig,
  background   = ST291_background,
  test         = "fisher",
  p_adjust     = "none",
  min_set_size = 1,
  max_set_size = 500,
  quiet        = TRUE
)

# Resultats ordenats i amb nom net

ORA_ST291_KEGG_sig <- ORA_ST291_KEGG %>%
  filter(p_value < 0.05) %>%
  arrange(p_adj) %>%
  mutate(set_name_clean = clean_pathway_name(set_name))

# Vista ràpida dels top 10
head(ORA_ST291_KEGG_sig, 10)
```

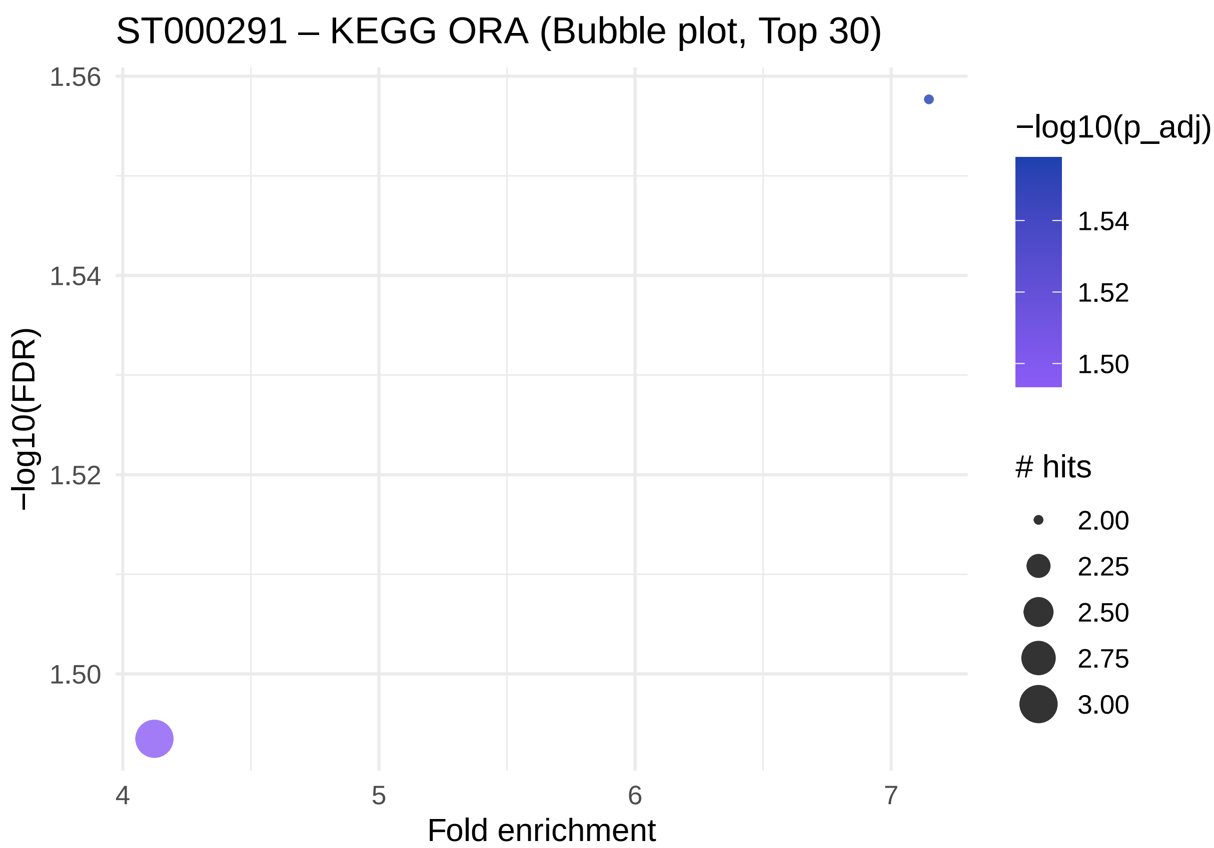
ORA Based on PubChem annotated results

```
## # A tibble: 2 x 9
##   set_id   set_name   n_selected_in_set n_in_set p_value overlap_features  p_adj
##   <chr>    <chr>                  <int>    <int>   <dbl> <chr>             <dbl>
## 1 hsa05208 Chemical ~                 2        5  0.0277 996;241          0.0277
## 2 hsa00350 Tyrosine ~                 3       13  0.0321 5816;996;780     0.0321
## # i 2 more variables: fold_enrichment <dbl>, set_name_clean <chr>
```

```r
## Barplot ORA - Top 20 pathways
top_ora <- ORA_ST291_KEGG_sig %>%
  slice(1:20) %>%
  mutate(set_name_clean = factor(set_name_clean, levels = rev(set_name_clean)))

ggplot(top_ora, aes(x = set_name_clean, y = -log10(p_adj))) +
  geom_col(fill = "#3B82F6") +
  coord_flip() +
  labs(
    title = "ST000291 - KEGG ORA (Top 20 pathways)",
    x     = "KEGG pathway",
    y     = "-log10(FDR)"
  ) +
  theme_minimal(base_size = 12)
```




```r
## Bubble plot ORA - Top 30
top_ora <- ORA_ST291_KEGG_sig %>%
  slice(1:30)

ggplot(top_ora, aes(
  x = fold_enrichment,
  y = -log10(p_adj),
  size = n_selected_in_set,
  color = -log10(p_adj)
)) +
  geom_point(alpha = 0.8) +
  scale_color_gradient(low = "#8B5CF6", high = "#1E40AF") +
  labs(
    title = "ST000291 - KEGG ORA (Bubble plot, Top 30)",
    x     = "Fold enrichment",
    y     = "-log10(FDR)",
    size  = "# hits"
  ) +
  theme_minimal(base_size = 12)
```

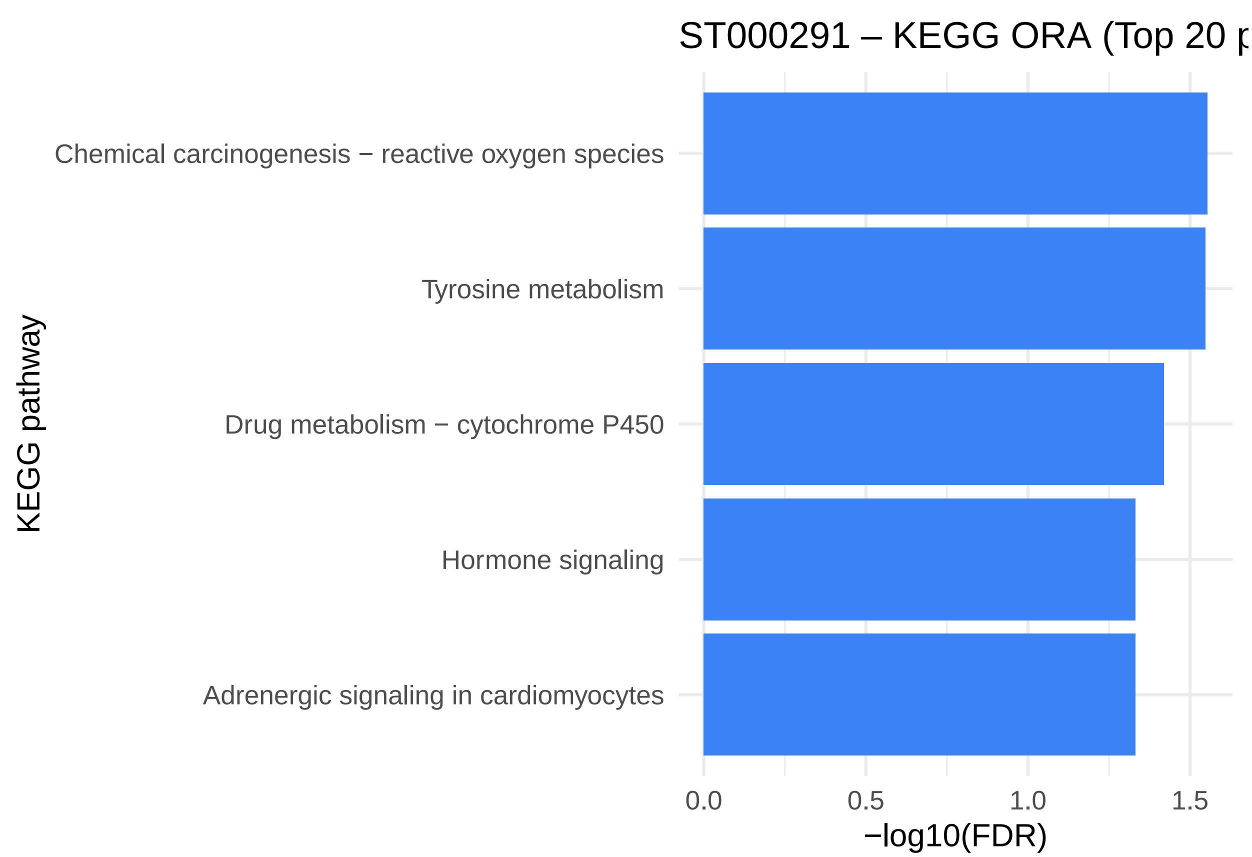



```r
## -------------------------------
## 1) ORA (Over-Representation Analysis)
## -------------------------------

ORA_ST291_HMDB_KEGG <- ora_test(
  eset          = KEGGset_HMDB,
  selected      = ST291_HMDB_sig,
  background    = ST291_HMDB_background,
  test          = "fisher",
  p_adjust      = "none",
  min_set_size = 1,
  max_set_size = 500,
  quiet         = TRUE
)

# Resultats ordenats i amb nom net

ORA_ST291_HMDB_KEGG_sig <- ORA_ST291_HMDB_KEGG %>%
  filter(p_value < 0.05) %>%
  arrange(p_adj) %>%
  mutate(set_name_clean = clean_pathway_name(set_name))

# Vista ràpida dels top 10
head(ORA_ST291_HMDB_KEGG_sig, 10)
```

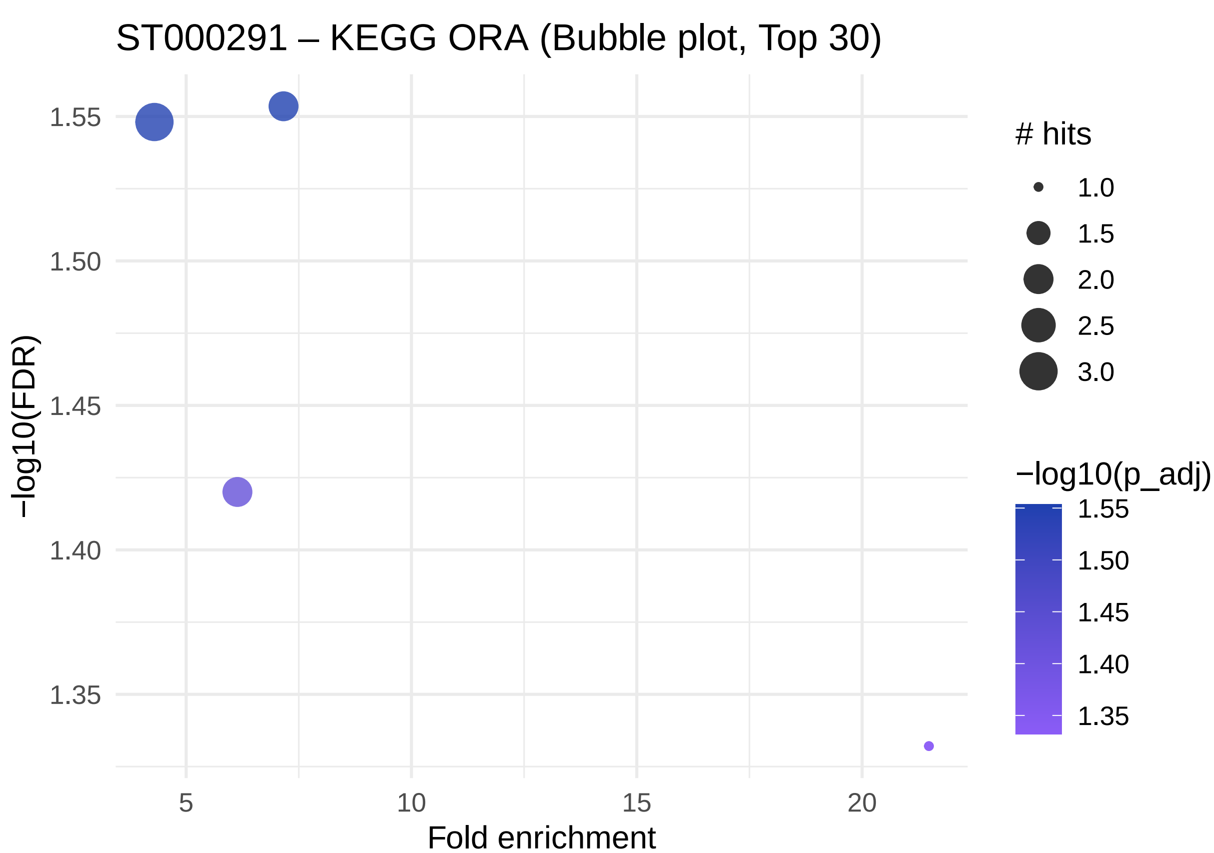
ORA based on HMDB annotated results

```
## # A tibble: 5 x 9
##   set_id   set_name   n_selected_in_set n_in_set p_value overlap_features  p_adj
##   <chr>    <chr>                  <int>    <int>   <dbl> <chr>             <dbl>
## 1 hsa05208 Chemical ~                 2        6  0.0280 HMDB0000228;HMD~ 0.0280
## 2 hsa00350 Tyrosine ~                 3       15  0.0283 HMDB0000068;HMD~ 0.0283
## 3 hsa00982 Drug meta~                 2        7  0.0380 HMDB0060686;HMD~ 0.0380
## 4 hsa04081 Hormone s~                 1        1  0.0465 HMDB0000068      0.0465
## 5 hsa04261 Adrenergi~                 1        1  0.0465 HMDB0000068      0.0465
## # i 2 more variables: fold_enrichment <dbl>, set_name_clean <chr>
```

```
## Barplot ORA - Top 20 pathways
top_ora <- ORA_ST291_HMDB_KEGG_sig %>%
  slice(1:20) %>%
  mutate(set_name_clean = factor(set_name_clean, levels = rev(set_name_clean)))

ggplot(top_ora, aes(x = set_name_clean, y = -log10(p_adj))) +
  geom_col(fill = "#3B82F6") +
  coord_flip() +
  labs(
    title = "ST000291 - KEGG ORA (Top 20 pathways)",
    x     = "KEGG pathway",
    y     = "-log10(FDR)"
  ) +
  theme_minimal(base_size = 12)
```



```r
## Bubble plot ORA - Top 30
top_ora <- ORA_ST291_HMDB_KEGG_sig %>%
  slice(1:30)

ggplot(top_ora, aes(
  x = fold_enrichment,
  y = -log10(p_adj),
  size = n_selected_in_set,
  color = -log10(p_adj)
)) +
  geom_point(alpha = 0.8) +
  scale_color_gradient(low = "#8B5CF6", high = "#1E40AF") +
  labs(
    title = "ST000291 - KEGG ORA (Bubble plot, Top 30)",
    x     = "Fold enrichment",
    y     = "-log10(FDR)",
    size  = "# hits"
  ) +
  theme_minimal(base_size = 12)
```



MSEA for the ST000291 dataset

```r
## --------------------------------
## 2) MSEA / QEA (Quantitative Enrichment Analysis)
## --------------------------------
```

```r
## ST291_scores ha de ser un vector numèric amb noms = PubChem IDs.
## Exemple d'estructura (NO s'executa):
##   ST291_scores <- setNames(topTable$logFC, topTable$PubChemCID)

# QEA/MSEA
```

```r
## ============================
## Construir ST291_scores
## ============================

# Comprovem que PubChemCID és caracter i no factor
topCranVSBase <- topCranVSBase %>%
  dplyr::mutate(PubChemCID = as.character(PubChemCID))

# Vector numeric amb noms = PubChem
ST291_scores <- setNames(topCranVSBase$log2FC, topCranVSBase$PubChemCID)
ST291_scores <- ST291_scores[!is.na(ST291_scores)]
str(ST291_scores)
```

```
##  Named num [1:1358] 2.92 3.27 2.18 2.41 3.18 ...
##  - attr(*, "names")= chr [1:1358] "54678503" "71485" "94214" "5378303" ...
```

```r
head(ST291_scores)
```

```
## 54678503    71485    94214  5378303     3037    10178
## 2.916890 3.267987 2.180373 2.414795 3.179669 3.331534
```

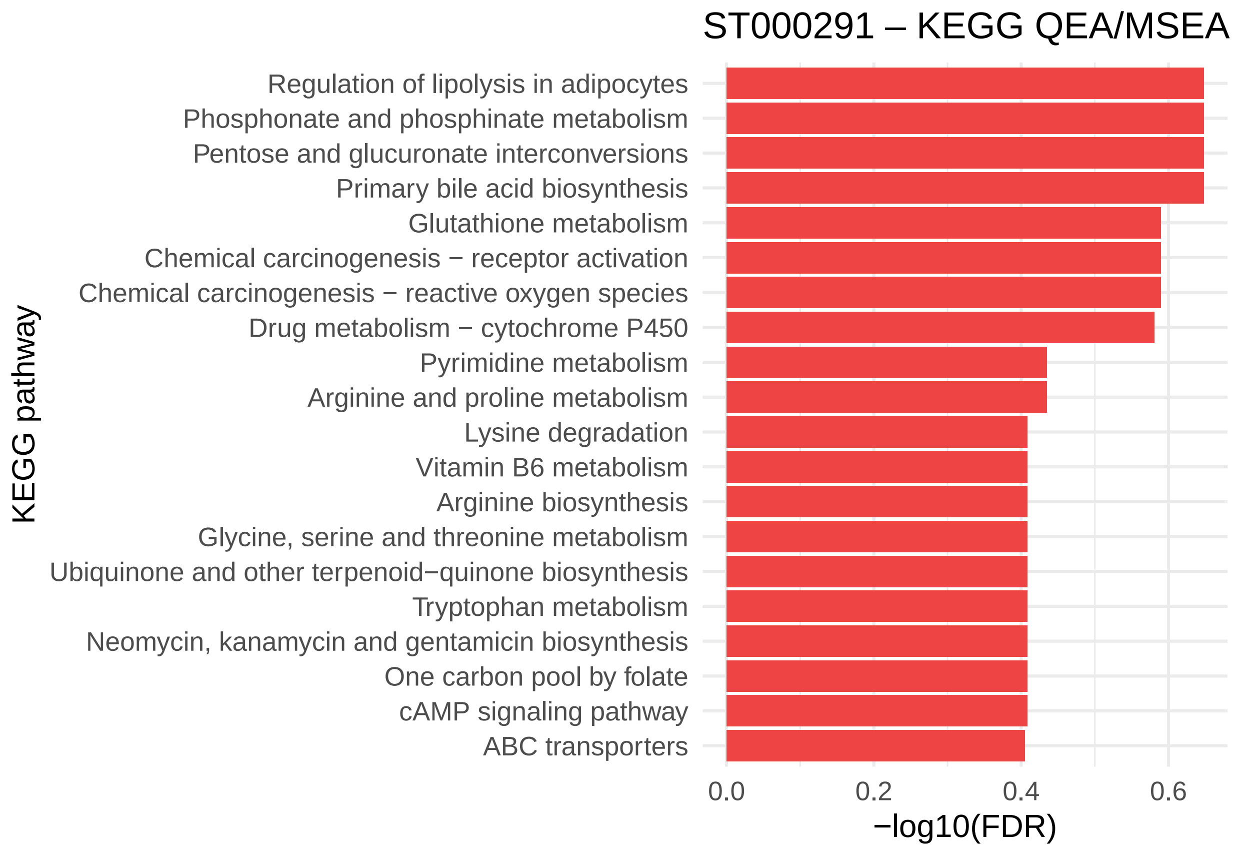
```r
stopifnot(is.numeric(ST291_scores))
stopifnot(length(ST291_scores) == length(names(ST291_scores)))
stopifnot(sum(is.na(ST291_scores)) == 0)
```

```r
# Comprovar estructura
QEA_ST291_KEGG <- qea_test(
  eset            = KEGGset_PubChem,
  scores          = ST291_scores,
  nperm           = 1000,
  p_adjust        = "BH",
  min_set_size    = 3,
  max_set_size    = 500,
  return_profiles = FALSE,
  quiet           = TRUE
)

QEA_ST291_table <- QEA_ST291_KEGG$table %>%
  filter(!is.na(p_value)) %>%
  arrange(p_adj) %>%
  mutate(set_name_clean = clean_pathway_name(set_name))

# Vista ràpida dels top 10 (QEA)
head(QEA_ST291_table, 10)
```

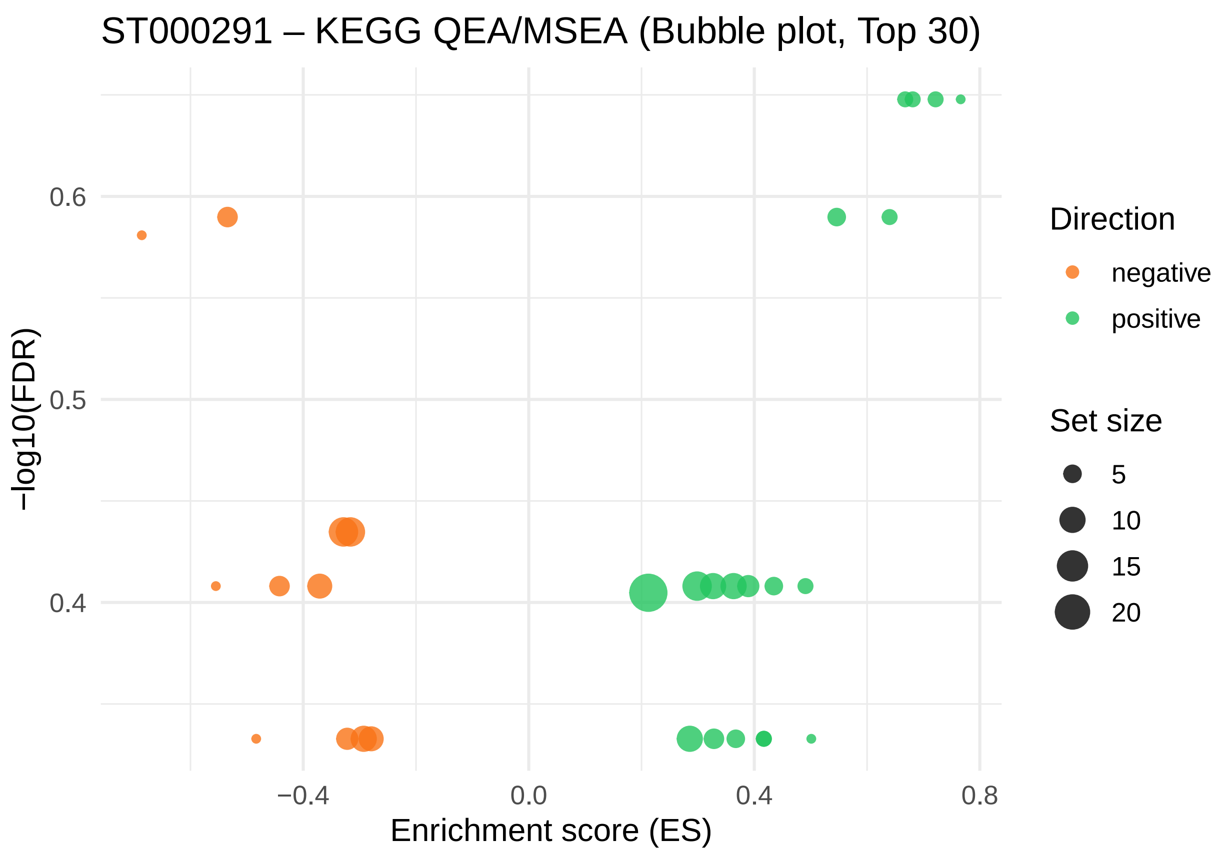
```
##      set_id
## 1  hsa04923
## 2  hsa00440
## 3  hsa00040
## 4  hsa00120
```

```
## 5  hsa00480
## 6  hsa05207
## 7  hsa05208
## 8  hsa00982
## 9  hsa00240
## 10 hsa00330
##                                                                         set_name
## 1              Regulation of lipolysis in adipocytes - Homo sapiens (human)
## 2             Phosphonate and phosphinate metabolism - Homo sapiens (human)
## 3           Pentose and glucuronate interconversions - Homo sapiens (human)
## 4                     Primary bile acid biosynthesis - Homo sapiens (human)
## 5                             Glutathione metabolism - Homo sapiens (human)
## 6       Chemical carcinogenesis - receptor activation - Homo sapiens (human)
## 7  Chemical carcinogenesis - reactive oxygen species - Homo sapiens (human)
## 8                  Drug metabolism - cytochrome P450 - Homo sapiens (human)
## 9                              Pyrimidine metabolism - Homo sapiens (human)
## 10                  Arginine and proline metabolism - Homo sapiens (human)
##    n_in_set         ES direction p_value     p_adj
## 1         4  0.7215657  positive   0.004 0.2250000
## 2         4  0.6809453  positive   0.008 0.2250000
## 3         4  0.6676514  positive   0.012 0.2250000
## 4         3  0.7660517  positive   0.012 0.2250000
## 5         6 -0.5342702  negative   0.021 0.2571429
## 6         4  0.6399557  positive   0.023 0.2571429
## 7         5  0.5461936  positive   0.024 0.2571429
## 8         3 -0.6863469  negative   0.028 0.2625000
## 9        13 -0.3163855  negative   0.047 0.3675000
## 10       13 -0.3286817  negative   0.049 0.3675000
##                                        set_name_clean
## 1              Regulation of lipolysis in adipocytes
## 2             Phosphonate and phosphinate metabolism
## 3           Pentose and glucuronate interconversions
## 4                     Primary bile acid biosynthesis
## 5                             Glutathione metabolism
## 6       Chemical carcinogenesis - receptor activation
## 7  Chemical carcinogenesis - reactive oxygen species
## 8                  Drug metabolism - cytochrome P450
## 9                              Pyrimidine metabolism
## 10                  Arginine and proline metabolism
```

```r
## Barplot QEA - Top 20 pathways
top20_qea <- QEA_ST291_table %>%
  slice(1:20) %>%
  mutate(set_name_clean = factor(set_name_clean, levels = rev(set_name_clean)))

ggplot(top20_qea, aes(x = set_name_clean, y = -log10(p_adj))) +
  geom_col(fill = "#EF4444") +
  coord_flip() +
  labs(
    title = "ST000291 - KEGG QEA/MSEA (Top 20 pathways)",
    x     = "KEGG pathway",
    y     = "-log10(FDR)"
  ) +
  theme_minimal(base_size = 12)
```




```r
## Bubble plot QEA - Top 30 (ES vs -log10(FDR))
top30_qea <- QEA_ST291_table %>%
  slice(1:30)

ggplot(top30_qea, aes(
  x = ES,
  y = -log10(p_adj),
  size = n_in_set,
  color = direction
)) +
  geom_point(alpha = 0.8) +
  scale_color_manual(values = c(positive = "#22C55E", negative = "#F97316")) +
  labs(
    title = "ST000291 - KEGG QEA/MSEA (Bubble plot, Top 30)",
    x     = "Enrichment score (ES)",
    y     = "-log10(FDR)",
    size  = "Set size",
    color = "Direction"
  ) +
  theme_minimal(base_size = 12)
```



ChemRich (outline)

ChemRich performs enrichment based on **chemical similarity clusters**, not predefined pathways. It typically requires:

- effect size (e.g. log2FC),
- p-values,
- a structural identifier (InChIKey, SMILES, etc.).

Since ChemRich is provided as a web tool, the usual workflow is:

1. Prepare a CSV with at least ID, log2FC and p-value.
2. Upload to ChemRich web.
3. Export the results table.
4. Re-import into R for comparison.

MetaboAnalyst Web (outline)

MetaboAnalyst Web is used here as a practical reference point ("gold standard"). The workflow is:

1. Prepare an input table in the expected format (ID + intensities or statistics).
2. Upload and run:
   - ORA (e.g. KEGG pathways),
   - MSEA (for full tables such as ST000291).
3. Export results as CSV and save under `results/MetaboAnalyst/`.

Comparison of results (skeleton)

The comparison strategy is:

1. Harmonize pathway identifiers across tools (e.g. KEGG pathway IDs).
2. Select top-N hits per method (by adjusted p-value).
3. Build tables of overlaps and, optionally, visual summaries.

Below is a simple skeleton for comparing ORA results for ST000291 using KEGG.